

“Evaluation of Two Approaches for Rainfall Prediction in the Junín Region: A Hybrid Classification-Regression Model and a Balanced Binary Classification Model”

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Abstract- This study addresses the challenge of rainfall prediction in the Junín region, which stems from the high proportion of records with no precipitation, introducing significant bias into traditional forecasting models. The main objective was to develop and evaluate two alternative approaches: a hybrid model combining binary classification to determine the occurrence of rain and regression to estimate its quantity; and a balanced binary classification model trained with resampling techniques to directly predict discrete precipitation levels. Although the hybrid model allows for a more detailed estimation of rainfall volume, results show that the balanced binary model achieved better overall performance, particularly in detecting significant rainfall events.

1. Introduction

Extreme rainfall in the Junín region is a critical climatic factor for the agricultural sector, particularly in relation to irrigation planning, crop management, and land productivity. During the rainy season, variability in the amount and intensity of precipitation directly affects water availability for agricultural use, potentially causing both water deficits and excesses that damage crops.

According to the National Institute of Civil Defense (INDECI), between 2012 and 2023, 136 flood emergencies were recorded in the Junín region, affecting more than 13,000 people, 2,200 homes, and 2,300 hectares of farmland, mainly in the districts of Chilca, Jauja, Concepción, and Huancayo. These extreme events not only threaten harvests but also hinder efficient irrigation management, highlighting the need for predictive tools that can anticipate the occurrence and magnitude of rainfall to support better agricultural decision-making.

Moreover, based on atmospheric and temporal data, it has been observed that approximately 95% of the records show zero values in the precipitation variable, which is expressed in terms of the vertical depth of water that would cover a horizontal surface of land in one hour, measured in millimeters (mm).

Given this scenario and the marked imbalance of zero-value records in the target precipitation variable, this study proposes the hypothesis that using two models with distinct approaches can mitigate the effects of imbalance and improve predictive capacity. To this end, two alternative approaches were developed and evaluated: a

hybrid model consisting of a binary classification stage followed by a regression to estimate rainfall quantity, and a balanced binary classification model designed to directly predict discrete precipitation levels. The main objective was to determine which of these approaches is more effective in anticipating the occurrence and intensity of rainfall. Ultimately, this study aims to contribute to better agricultural management in vulnerable regions such as Junín, where accurate rainfall prediction is key to optimizing irrigation, preventing crop losses, and making informed decisions regarding water resource use.

2. Related Work

Below is a summary of scientific articles related to the objective of this study.

Title: Monthly Precipitation Prediction Using Artificial Neural Networks for the Cali River Basin, Colombia

This study applies Artificial Neural Networks to predict monthly precipitation in the Cali River basin under RCP climate scenarios. Data from 35 stations (1972–2016) were used, with missing values completed using Multilayer Perceptron networks. The networks were trained with hyperbolic functions and up to 6000 iterations, achieving up to 98% accuracy, proving useful for local planning.

Title: MT-HCCAR: Multi-Task Deep Learning with Hierarchical Classification and Attention-based Regression for Cloud Property Retrieval



This article proposes a deep learning multitask model called MT-HCCAR, which simultaneously addresses hierarchical cloud classification and cloud optical thickness (COT) estimation. The model consists of two main subnetworks: a hierarchical classification subnetwork that first detects cloud presence and then identifies its phase (liquid, solid, or mixed), and an attention-assisted regression subnetwork that estimates COT only when a cloud is detected. The latter incorporates a cross-attention mechanism that integrates information from the classification subnetwork to enhance regression accuracy. Both subnetworks share a base convolutional architecture and are jointly trained using a multitask optimization scheme, leveraging task synergy to improve overall performance. Training was conducted using simulated satellite sensor data (OCI, VIIRS, and ABI), with K-fold cross-validation and the one standard error rule (ISE) for final model selection.

Title: A Methodology Based on Random Forest to Estimate Precipitation Return Periods: A Comparative Analysis with Probability Density Functions in Arequipa, Peru.

This study compares Random Forest (RF) with probability density functions to estimate extreme precipitation events in Arequipa, Peru. Using data from 26 stations (1965–2013), RF showed higher accuracy in 61–69% of cases compared to PDFs such as Gumbel or GEV. RF stood out for its robustness against noisy data and lower error in return periods ranging from 2 to 100 years.

Title: Climate Variability and Its Impact on Food Security Crop Yields in the Highlands of Junín

This study analyzes how climate variability (temperature, precipitation, and humidity) affects the yield of 27 crops in the highlands of Junín between 2015 and 2022. Climate and yield data were obtained from MIDAGRI and SENAMHI, and adaptability strategies (technical and agroecological) applied by farmers were evaluated. Statistical methods such as t-tests, Pearson correlation, and binary logistic regression were used for analysis.

Title: Advancing preeclampsia prediction: a tailored machine learning pipeline integrating resampling and ensemble models for handling imbalanced medical data

The study developed a specialized pipeline for early prediction of preeclampsia in imbalanced medical datasets. Eight resampling techniques were applied, including SMOTE, ADASYN, and the novel IWGMM, varying the minority/majority ratio (MMR) between 0.05 and 1. Subsequently, models were trained using six ensemble learning algorithms (e.g., GBDT and Random Forest), generating 4,608

configurations. Performance evaluation used metrics adapted to imbalance, including G-mean, MCC, AP, and AUC. An iterative optimization process adjusted resampling, algorithm, and MMR in the most effective order, and the pipeline's generalization capacity was validated on three additional public datasets.

Title: SMOTE for Handling imbalanced Data Problem: A Review

This article addresses the issue of class imbalance in classification tasks, where one class is represented by significantly fewer instances than others, impairing model accuracy—especially for the minority class. As a solution, it highlights SMOTE (Synthetic Minority Oversampling Technique), a pioneering oversampling method that generates synthetic instances by combining a minority instance with its K nearest neighbors in feature space. This helps avoid overfitting and improves the classifier's ability to find more precise decision boundaries between classes.

3. Methodology

■ Proposed Approach(es):

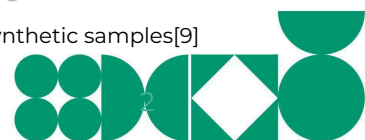
The dataset used in this study contains approximately 95% zero values in the target variable. This strong imbalance introduces significant bias into the model, as it tends to favor the majority class. To mitigate this issue and improve predictive performance, the following approaches were proposed:

- A. **Hybrid Model:** This approach consists of an initial binary classification stage that predicts the occurrence of precipitation, followed by a regression model that estimates the amount of rainfall in millimeters. The regression model is activated only when the classifier predicts a positive rainfall event.
- B. **Balanced Binary Classification Model:** This approach transforms the target variable into two discrete categories ("No Rain/Light" (level 0, less than 2.5 mm), and "Moderate/Heavy" (level 1, greater than or equal to 2.5 mm)). The model is then trained using the SMOTE[9] technique, which balances the classes by generating synthetic data for the minority category.

The formula for generating synthetic samples using the SMOTE technique is as follows:

$$x_{new} = x_i + \delta \cdot (x_{neighbor} - x_i)$$

Figure 1: Generation formula for synthetic samples[9]



Where:

x_i = A sample from the minority class. In this case, it corresponds to a moderate/heavy rainfall instance.

$x_{neighbor}$ = Another minority class sample located near x_i .

x_{new} = Synthetic sample generated for the minority class.

δ = A random number between 0 and 1.

A. Data Acquisition.

The information used to train the models was obtained from datasets available on the National Open Data Platform of the Government of Peru. These datasets originate from the Automatic Meteorological Station (EMA) and the BSRN Station for Solar-Earth Energy Balance. In total, these two datasets served as the foundation for the model training process and contain the following attributes:

VARIABLE	DESCRIPTION	TYPE
FECHA_CORTE	Data cutoff date	Numeric
UBIGEO	Geographical Location Code denoted as 'DDppdd' (Department, Province, District)	Alphanumeric
YY	Year, año de registro	Numeric
MM	Month, mes de registro	Numeric
DD	Day, día de registro	Numeric
HH	Hour, hora promedio del registro	Numeric
TT	Air Temperature 2 meters (temperatura del aire a 2 metros) - [°C]	Numeric
HR	Relative Humidity (Humedad Relativa) - [%]	Numeric
RR	Precipitation (precipitación) - [mm]	Numeric
PP	Atmospheric pressure (presión atmosférica) - [hPa]	Numeric
FF	Wind speed (velocidad del aire) - [m/s]	Numeric
DD	Wind direction (dirección del aire) - [degrees]	Numeric

Figure 2: Data Dictionary of the EMA Dataset

VARIABLE	DESCRIPTION	TYPE
FECHA_CORTE	Data cutoff date	Numeric
UBIGEO	Geographical Location Code denoted as 'DDppdd' (Department, Province, District)	Alphanumeric
year	Year, año de registro	Numeric
month	Month, mes de registro	Numeric
day	Day, día de registro	Numeric
hour	Hour, hora promedio del registro	Numeric
incide_LW_rad	Incident longwave irradiance (irradiancia incidente de onda larga) W m ⁻²	Numeric
emite_LW_rad	Emitted longwave irradiance (irradiancia emitida de onda larga) W m ⁻²	Numeric
direct_SW_rad	Direct shortwave irradiance (irradiancia directa de onda corta) W m ⁻²	Numeric
global_SW_rad	Global shortwave irradiance (irradiancia global de onda corta) W m ⁻²	Numeric
diffus_SW_rad	Diffuse shortwave irradiance (irradiancia difusa de onda corta) W m ⁻²	Numeric
reflec_SW_rad	Reflected shortwave irradiance (irradiancia reflejada de onda corta) W m ⁻²	Numeric

Figure 3: Data Dictionary of the BSRN Dataset

Subsequently, the columns for year, month, day, and hour in the EMA dataset were renamed to match the corresponding column names in the BSRN dataset, in order to merge the columns from both datasets into a single unified dataset. After this integration, a one-hour forward shift was applied to the variable RR, ensuring that each recorded value corresponds to the precipitation observed one hour after the respective timestamp.

B. Data Preprocessing.

First, rows containing null values were removed, as the proportion of missing data in any column did not exceed 25%. Next, the months with the highest frequency of rainfall were identified in order to reduce the proportion of records with zero values in the target variable and focus the analysis on predicting rainfall during rainy periods. Additionally, the "hour" variable was transformed into two radial variables due to its cyclical nature, and the remaining temporal variables were discarded. Principal Component Analysis (PCA) was then applied to project the variables onto PC1 and PC2, allowing for an assessment of the variability contributed by each variable in its two projections. Subsequently, a correlation matrix was generated to identify redundant columns. The redundant columns were removed from the dataset, retaining only those considered most representative of the underlying characteristics.

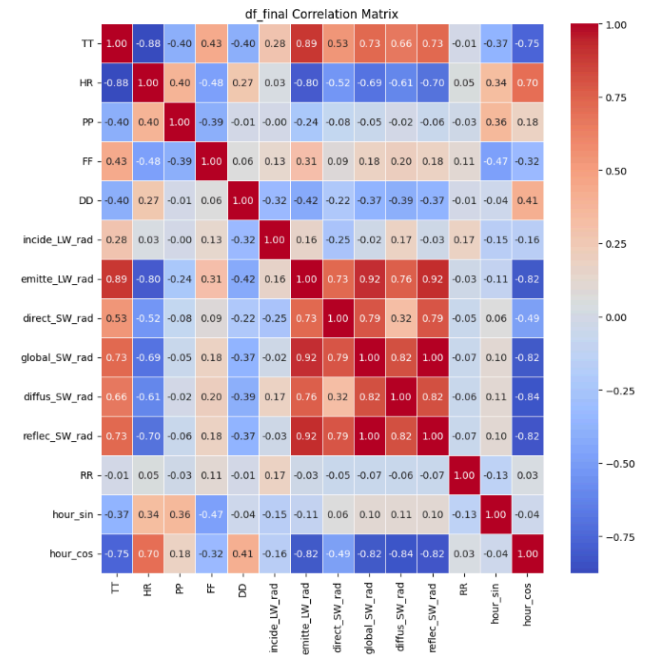


Figure 4: Correlation Matrix

C. Model Training and Optimal Model Selection

In both approaches, the dataset was divided into two subsets using the train-test split technique [3], allocating 80% of the data to the training set and 20% to the test set.

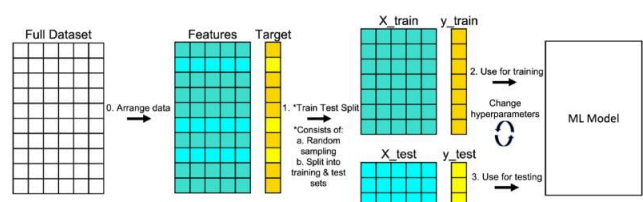


Figure 5: Train-test split

Source: Galarnyk, M. (s. f.). Train Test Split: What It Means and How to Use It

To ensure a robust evaluation and reduce variance in the results, 10-fold cross-validation was employed. This technique enabled a more reliable estimation of model performance across different subsets of the training set, mitigating the risk of overfitting and promoting better generalization capacity

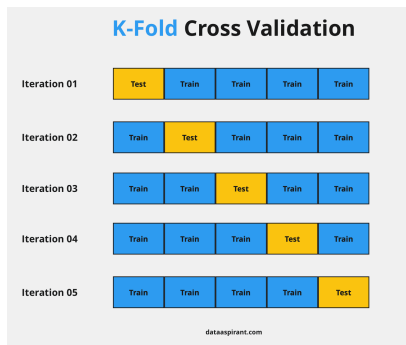


Figure 6: K-Fold Cross-Validation

Source: Arif, A. (2020). How cross-validation works in machine learning

Hybrid Model:

In the first stage of this approach, which focuses on classifying the occurrence of precipitation, various machine learning models were evaluated to identify the one with the best performance. These included logistic regression, K-Nearest Neighbors classification, and Decision Tree classification. The primary evaluation metric was balanced accuracy, and the model with the highest score was selected.

In the regression stage, algorithms such as linear regression, K-Nearest Neighbors regression, and Decision Tree regression were assessed. Pipelines incorporating feature standardization were applied throughout.

During implementation, the classification model was trained first. It received the feature vector as input and predicted the probability of precipitation occurrence. If the prediction was positive, the same feature vector was passed to the regression model to estimate the accumulated rainfall. Otherwise, the output value was automatically set to 0 mm, thereby avoiding unnecessary predictions under dry conditions.

Subsequently, the individual performance of each model and the combined system was evaluated. Balanced accuracy was used for the classification model, while mean squared error (MSE), mean absolute error (MAE), and the coefficient of determination (R^2) were calculated for both the regression model and the integrated system.

Balanced Binary Classification Model:

In this approach, manual undersampling was first applied to reduce the majority class in the training set and prevent an excess of synthetic samples. Subsequently, the SMOTE technique was used to balance the classes, resulting in a more equitable class distribution.

Once the class balancing process was completed, model training and optimal model selection were carried out using a procedure similar to that of the first approach.

4. Experimentation and Results

Experimental Setup:

The models were developed in Python, using Google Colab as the development environment.

After completing the data preprocessing stage, each approach was implemented as follows:

A. Hybrid Model:

The first stage of the hybrid machine learning model focuses on regression to predict the amount of precipitation. Among the various algorithms tested, the Ridge regression algorithm yielded the best results due to its low mean squared error and minimal variance. This was further enhanced by incorporating pipelines with the StandardScaler for feature standardization

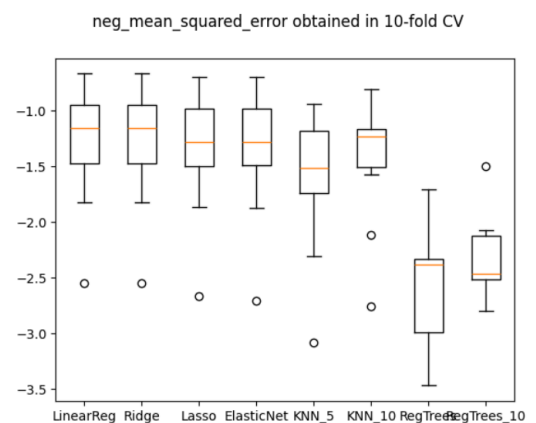


Figure 7: Results of Algorithms Used with Pipelines in the Hybrid Model

Mean squared error: 1.8499222827929134
 Mean absolute error: 0.7659956843832113
 Explained variance score: -0.02380253434218149
 R^2 score: -0.026137900402035452

Figure 8: Regression Stage Results Using the Ridge Algorithm in the Hybrid Model



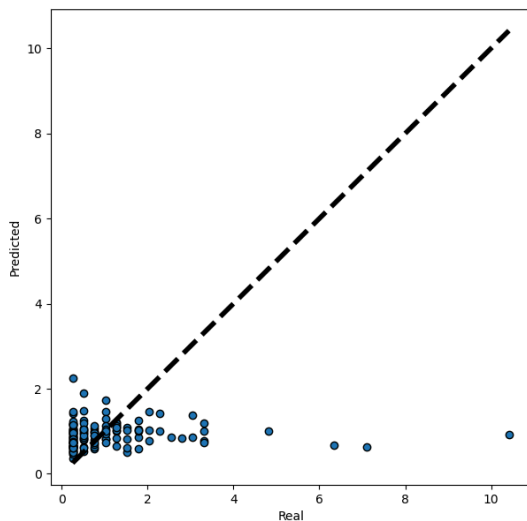


Figure 9: Scatter Plot – Comparison Between Predicted and Actual Values

For the classification stage, the aforementioned algorithms were applied in order to gain a clear understanding of which would yield the best possible performance for subsequent implementation in the hybrid model.

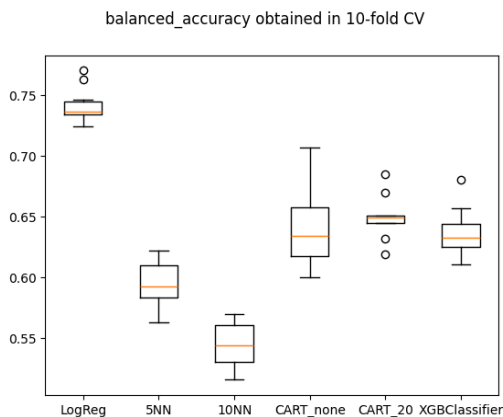


Figure 10: Boxplot of Metrics for Classification Algorithms

B. Balanced Binary Classification Model:

Class balancing was performed by first applying manual undersampling to reduce the majority class in the training set. This was followed by the application of the SMOTE technique, resulting in the following data distribution:

Data distribution before SMOTE

```
Counter({0: 3700, 1: 58})
```

Data distribution after SMOTE

```
Counter({0: 3700, 1: 3700})
```

Figure 11: Data Distribution Before and After Balancing

The target variable was divided into two classes based on rainfall levels: “No Rain/Light” (level 0) and “Moderate/Heavy” (level 1). The model was then trained using various algorithms, following the same procedure as in the first approach, in order to select the one with the best performance. The following results were obtained:

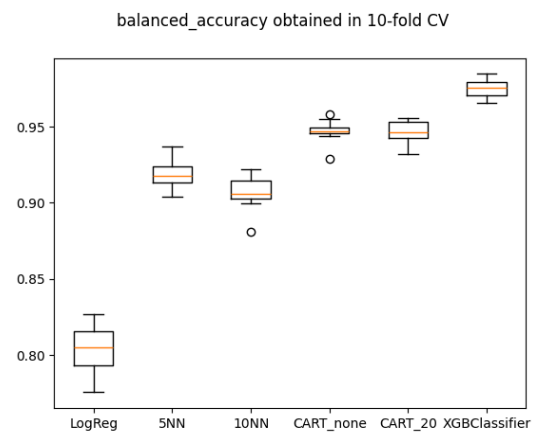


Figure 12: Boxplot of Metrics for the Algorithms Used During Training

Results and Discussion:

A. Hybrid Model:

Based on the results obtained from the algorithm selection process for each stage of the first approach, the DecisionTreeClassifier (max_depth=20) was chosen over LogisticRegression for the classification stage. When combined with Ridge regression in the proposed hybrid framework, this configuration yielded superior results according to the evaluation metrics Mean Squared Error (MSE), Mean Absolute Error (MAE), and the coefficient of determination (R^2) as described in the methodology.


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Classification:
Balanced Accuracy: 0.7520129202607012
Train set size: 498
Test set size: 765

Regression (applied only when rain is predicted):
MAE: 0.7422200376286493
MSE: 0.8181143067848059
R2: -0.5339591492348366

Classification + regression:
MAE: 0.2999133809406069
MSE: 0.3560385380001877
R2: -0.38737458018210247

```

Figure 13: Metric Results After Training the Hybrid Model (LogisticRegression)

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Classification:
Balanced Accuracy: 0.6503741448676403
Train set size: 498
Test set size: 239

Regression (applied only when rain is predicted):
MAE: 0.7440949947130745
MSE: 1.092349402135498
R2: -0.13876453552153256

Classification + regression:
MAE: 0.1486188048560722
MSE: 0.2668039536710145
R2: -0.03965437363708557

```

Figure 14: Metric Results After Training the Hybrid Model (DecisionTreeClassifier)

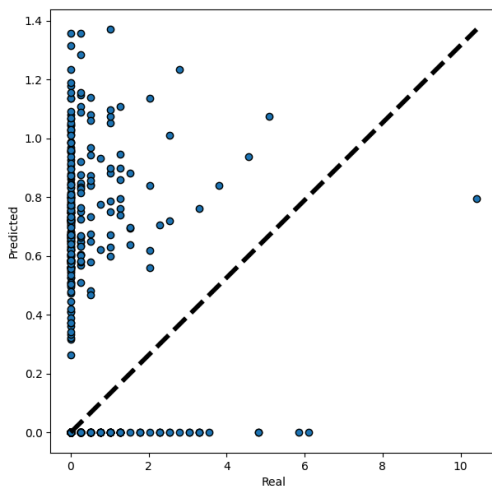


Figure 15: Scatter Plot – Hybrid Model Results, Comparison Between Predicted and Actual Values (DecisionTreeClassifier)

The result obtained from the model was not as expected, as it showed poor performance in predicting precipitation.

B. Balanced Binary Classification Model:

The LogisticRegression model was selected due to its higher balanced accuracy and its outstanding ability to detect moderate or heavy rainfall (recall = 0.93). This capability is a priority, as minimizing false negatives in that class is more critical than avoiding false alarms. The results obtained are shown below:

Balanced Accuracy score: 0.8057445200302342

Confusion Matrix:
[[998 325]
[2 12]]

Classification Report on the Validation Set:

	precision	recall	f1-score	support
0	1.00	0.75	0.86	1323
1	0.04	0.86	0.07	14
accuracy			0.76	1337
macro avg	0.52	0.81	0.46	1337
weighted avg	0.99	0.76	0.85	1337

Figure 16: Confusion Matrix and Classification Report - Balanced Binary Classification Model

As illustrated in Figure 16, the outcomes of the second approach demonstrated better performance in rainfall prediction compared to the hybrid model.

5. Conclusion

In summary, this study demonstrated that, initially, the hybrid approach yielded better results when using the DecisionTreeClassifier (max_depth=20) algorithm for the classification stage and the Ridge algorithm for the linear regression stage. However, it was observed that this approach delivered results below expectations and, in fact, could be categorized as poor.

For this reason, a second approach was adopted: the balanced binary classification model trained using the SMOTE technique, which enables the classifier to better learn the patterns associated with rainfall occurrence.

Ultimately, the best results were achieved using the LogisticRegression algorithm, as it most accurately predicted cases of moderate or heavy rainfall. Therefore, we conclude that the second approach provides the most optimal results.

6. Suggestions for Future Work

Improvement of Data Preprocessing:

- Implement advanced imputation techniques to handle missing or erroneous values in meteorological variables.
- Incorporate satellite or radar-based meteorological data sources to enhance the quality and spatial resolution of the dataset.

Evaluation of Deep Learning Models:

- Explore architectures such as LSTM, GRU, or Transformers to model temporal sequences and capture complex precipitation patterns.
- Compare their performance against the hybrid and balanced binary approaches.

Validation in Other Agricultural Regions of Peru:

- Replicate the approach in other regions with distinct agroclimatic characteristics, such as Arequipa, Cusco, or Piura, to assess its generalizability.

Model Explainability and Trust:

- Apply tools such as SHAP or LIME to interpret model decisions, which is crucial for adoption by farmers and institutions.

Multi-Horizon Precipitation Forecasting:

- Develop models capable of predicting not only hourly precipitation but also short- and medium-term rainfall forecasts.

7. Ethical Implications

Responsible Use of Data:

- It is essential to ensure principles such as data quality and privacy in meteorological datasets. Although these are not personal data, poor processing can lead to misguided agricultural decisions. To address this issue, it is crucial to implement automatic data validation processes, real-time quality control, and auditing mechanisms that guarantee the integrity and reliability of the information used by predictive models.

Risk of Blind Trust in Automated Models:

- If farmers or institutions rely solely on predictions without understanding their limitations or error margins, there is a risk of making inappropriate decisions that could harm crops. To prevent this, training in model interpretation should be promoted, and uncertainty reports or error margins should be required alongside predictions.

Responsibility and Transparency in Predictions:

- Models must be explainable and auditable to avoid arbitrary or unjustified decisions in resource allocation. This requires the use of interpretability techniques such as SHAP or LIME, maintaining detailed records of training and validation processes, and documenting model decisions to facilitate review by experts and stakeholders.

Ethics of Bias and Representativeness:

- If the model is trained on limited or non-representative data from the region, it may introduce biases that disproportionately affect certain areas or crop types. Ethically, such biases must be identified and corrected. This involves conducting fairness analyses of model performance across different subregions or

crop types and supplementing the data with more diverse sources.

Long-Term Sustainability of the System:

- If the system is implemented without a sustainability plan, it may fail over time, creating dependency on a technology that eventually becomes obsolete. Ethical design requires consideration of long-term impact and local capacity to maintain the tool. This includes incorporating a maintenance plan from the outset, providing ongoing training for local users, using open and scalable technologies, and integrating with existing technical capabilities in the region.

8. Project Repository Link

<https://github.com/MrLinoP/Modelos-Predictivos-Precipitacion>

9. Author Contribution Statement

Lino, J.: Formulation of the hypothesis, dataset acquisition, development of the methodology and code (first approach), and translation of the original Spanish version of the document into English.

Calderón, C.: Drafting of the methodology, hypothesis, objective, and proposed approach. Synthesis of two related works. Development of the methodology and code (second approach).

Chavarria, M.: Addition of two academic sources for the related work section. Contribution of methodological insights. Drafting of the experimentation and results, conclusions, future work suggestions, and ethical implications.

Izarra, L.: Synthesis of one related work. Contribution during the early stages of methodology development. Drafting of the experimentation and results, conclusions, future work suggestions, and ethical implications.

Mascco, P.: Acquisition of academic sources used as guidance for the model, drafting of the introduction, and contribution of ideas to the code. Drafting of the experimentation and results, conclusions, future work suggestions, and ethical implications.

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handling imbalanced medical data. *BioData Mining*, 18(25), 1–16. <https://doi.org/10.1186/s13040-025-00440-1>

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